

abstract

Stellar streams from tidally stripped globular clusters are sensitive to dark matter distribution across scales, from galaxy halos to small-scale structure. We expect globular cluster streams to exist in a variety of host galaxy types, but so far, they have only been observed in the Milky Way. We present evidence for the first extragalactic globular cluster stellar stream, identified in deep Hubble Space Telescope imaging of the ultra-diffuse galaxy, UGC9050-Dw1. The stream's morphology, colour, surface brightness, and apparent association with a compact source support the globular cluster progenitor interpretation observationally. We use generative stream modelling to present the first stream-based halo constraint for an ultra-diffuse galaxy and place limits on the inner density profile. Our dynamical bounds also point to a globular cluster progenitor, and provide halo mass estimates consistent with globular cluster counts. Extending the reach of globular cluster stream analysis to external galaxies opens a new chapter in dark matter science.

Recent capabilities to detect low-surface-brightness objects have led to the discovery of numerous ultra-diffuse galaxies [1, 2]; galaxies with dwarf-like stellar masses but Milky Way-like extents [3]. The dark matter content of ultra-diffuse galaxies are the subject of considerable debate [4–8]. The methods available for inferring their detailed mass and density distribution remain limited, since their characteristic low surface brightness means that measuring velocity dispersions or rotation curves is challenging [9]. A possible technique not dependent on tracer kinematics is the analysis of stellar streams [10]. Stellar streams form from the gravitational interplay between a galaxy's halo and a bound stellar system orbiting it. Tidal forces strip stars from the progenitor system into arms along the orbit, the dynamics and morphology of which are largely influenced by the host galaxy's mass distribution. In the Milky Way, thin and dynamically cold streams of stars stripped from globular clusters are particularly sensitive probes of the dark matter halo and substructure [11–14]. Here we present evidence for the first extragalactic stellar stream from a globular cluster progenitor and use it to constrain the dark matter halo of its ultra-diffuse host galaxy. We find that the progenitor mass is in the range expected for a globular cluster, and reproduce the observed brightness with simulated stellar populations. Our dynamical modelling suggests a massive dark matter host halo and constrains its inner density slope. To date, no extragalactic stellar streams from globular cluster progenitors have been confirmed, and previous candidates have been shown to be physically implausible [15–17] or simply not detected at a statistically significant level above the background stellar halo, even in nearby galaxies such as M31 [18]. With these results, we present a new, independent probe of the central density distribution in low-mass galaxies. By extending stream-based analyses to faint, low surface brightness systems, this work provides a new benchmark for testing competing models of dark matter.